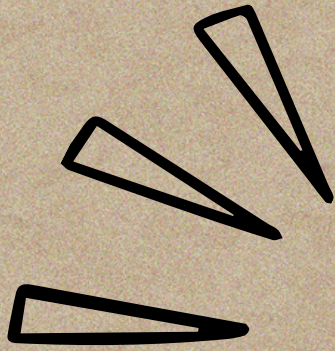
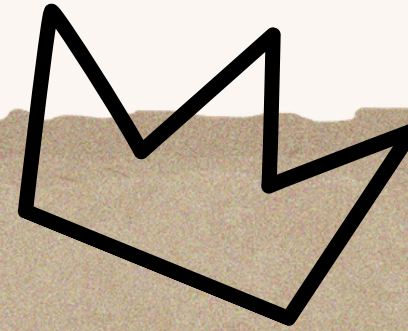
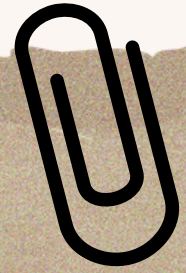
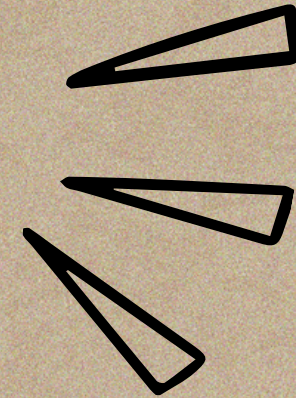


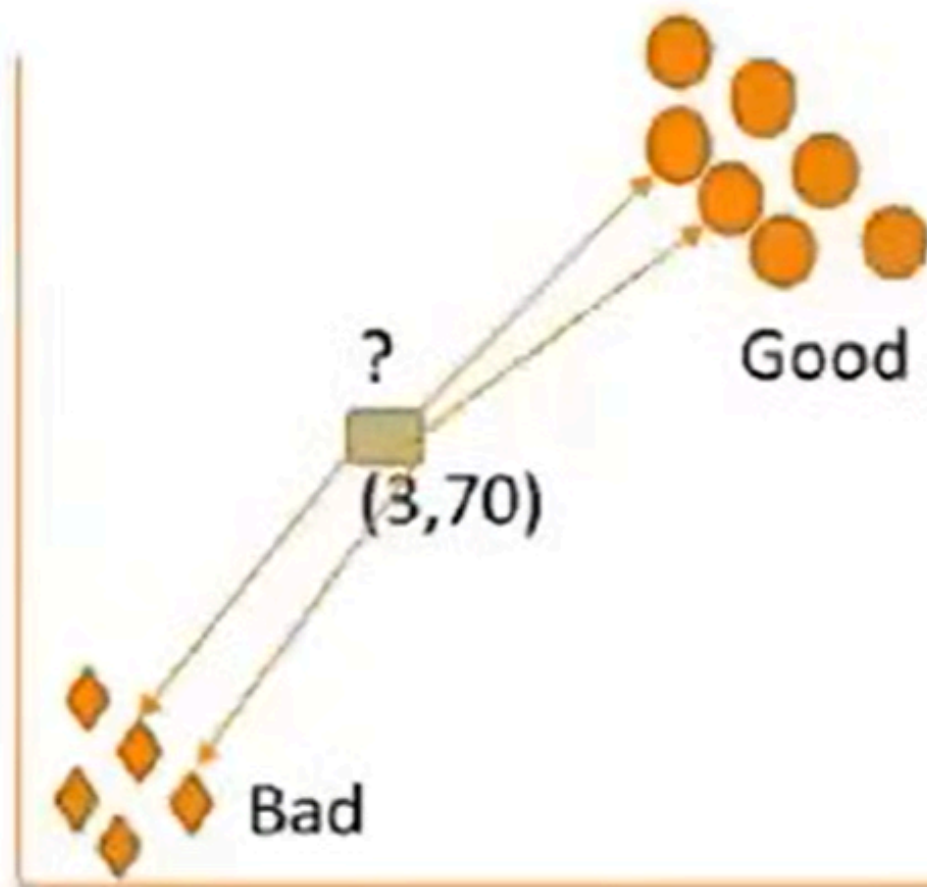


K Nearest Neighbours Algorithm



Classification Using KNN

K=4

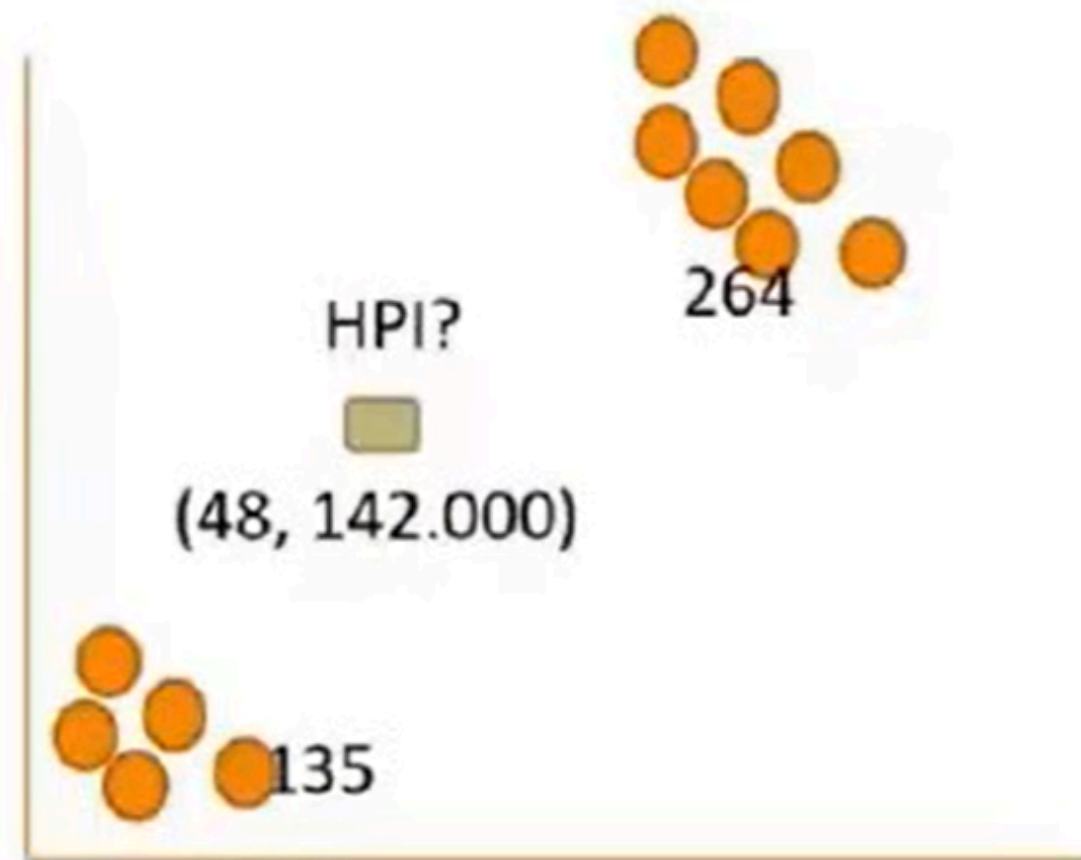


Name	Cigarettes	Weight	Heart Attack	Distance	
A	7	70	Bad	$\sqrt{(3-7)^2 + (70-70)^2} = 4$	①
B	7	40	Bad	$\sqrt{(3-7)^2 + (70-40)^2} = 30.27$	④
C	3	40	Good	$\sqrt{(3-3)^2 + (70-40)^2} = 30.00$	②
D	1	40	Good	$\sqrt{(3-1)^2 + (70-40)^2} = 30.07$	③
E	3	70	Bad		

Regression Using KNN

قيم عددية

K=3



Age	Loan	House Price Index	Distance
25	\$40,000	135	102000
35	\$60,000	256	82000
45	\$80,000	231	62000
20	\$20,000	267	122000
35	\$120,000	139	22000
52	\$18,000	150	124000
23	\$95,000	127	47000
40	\$62,000	216	80000
60	\$100,000	139	42000
48	\$220,000	250	78000
33	\$150,000	264	8000
48	\$142,000	?	

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Explanation

K=1

- using the training set to classify an unknown case
- (Age=33 and Loan=\$150,000) {Euclidean distance}.
- If K=1 then the nearest neighbor is the last case in the set
- with HPI=264.
- $D = \text{Sqrt}[(48-33)^2 + (142000-150000)^2] = 8000.01 \Rightarrow \text{HPI} = 264$

K=3

- the prediction for HPI is equal to the average of HPI for the top three neighbors
- $\text{HPI} = (264+139+139)/3 = 180.7$

TRY when K =4? What is HPI for it ?

Age	Loan	House Price Index	Distance
25	\$40,000	135	102000
35	\$60,000	256	82000
45	\$80,000	231	62000
20	\$20,000	267	122000
35	\$120,000	139	22000
52	\$18,000	150	124000
23	\$95,000	127	47000
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60	\$100,000	139	42000
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33	\$150,000	264	8000
48	\$142,000	?	

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

KNN pros and cons

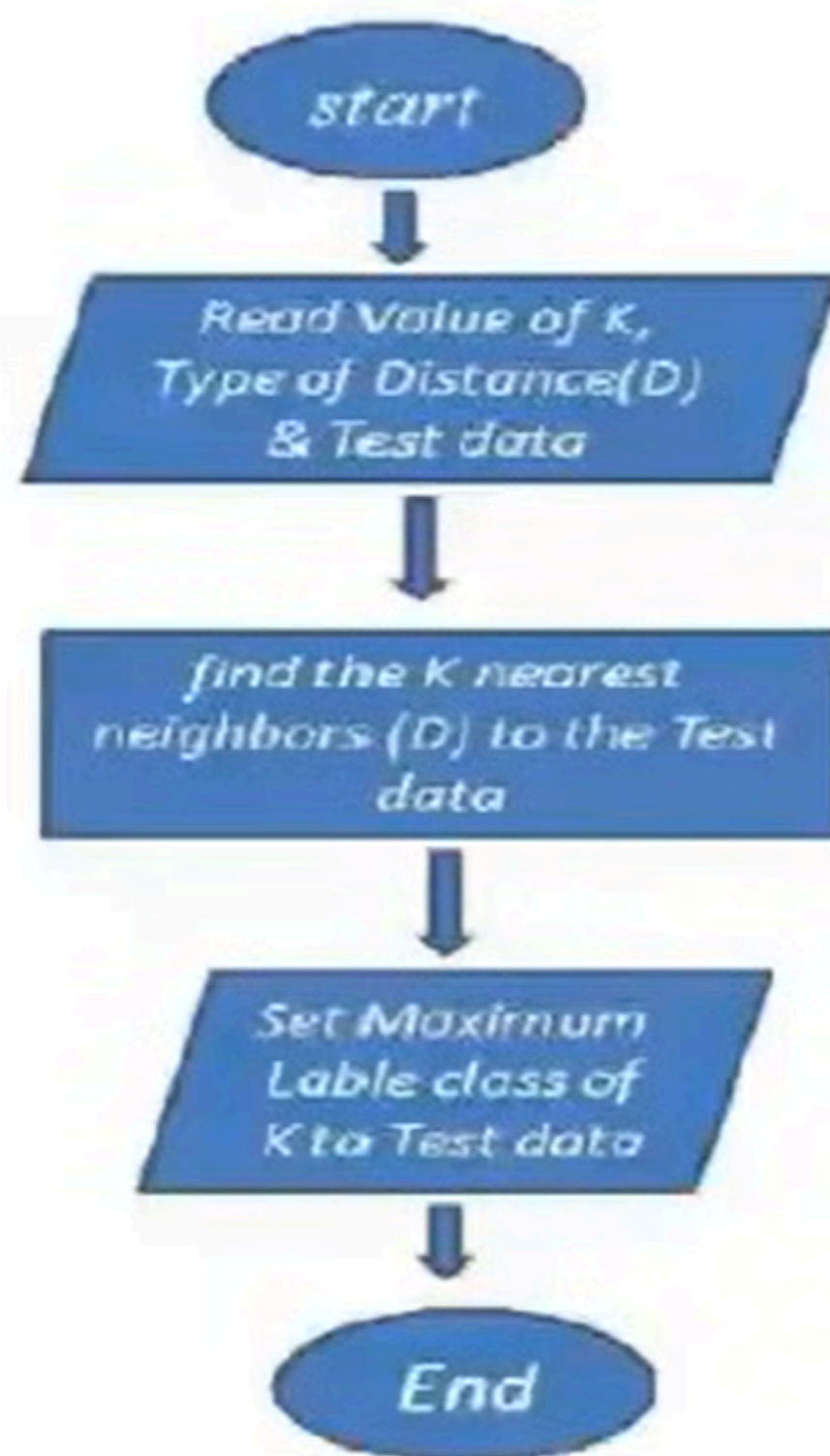
Pros

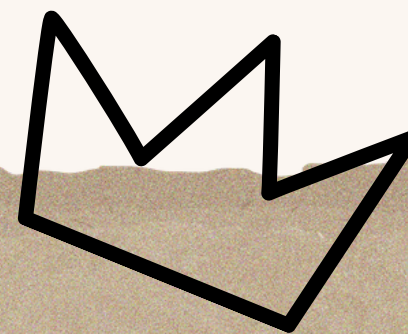
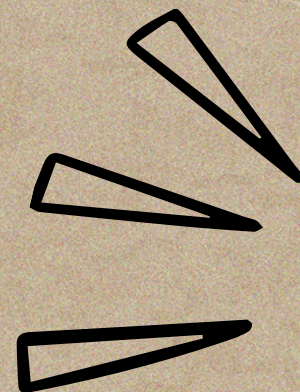
1. No Training Period: KNN is called Lazy Learner (Instance based learning).
2. new data can be added seamlessly which will not impact the accuracy of the algorithm.
3. KNN is very easy to implement, There are only two parameters required to implement KNN i.e. the value of K and the distance function (e.g. Euclidean or Manhattan etc.)

Cons

1. Does not work well with large dataset:
2. Does not work well with high dimensions
3. Need feature scaling: We need to do feature scaling (standardization and normalization)
4. Sensitive to noisy data, missing values.

Flow chart for KNN





**Thank
you**

